Gopalakrishnan Thirunellai Venkitachalam

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EDUCATION

Carnegie Mellon University (CMU) (GPA 4.0/4.0) Pittsburgh, PA
MS in Mechanical Engineering | Specialization: **Robotics, Artificial Intelligence & Motion Planning** May 2026
Relevant Courses: Planning & Decision-Making for Robotics, Generative AI, Advanced NLP, 11-785: Deep Learning

Indian Institute of Technology Madras (IIT Madras) (GPA 8.71/10.0) Chennai, India
B.Tech (**Honors**) in Mechanical Engineering | **Minor** in **Artificial Intelligence and Machine Learning** June 2024
Relevant Courses: Machine Learning, Reinforcement Learning, Robotics, Multi-Armed Bandits, Stochastic Processes

EXPERIENCE

Search Based Planning Lab, Robotics Institute, CMU Pittsburgh, PA
Graduate Research Assistant | Advisor: Prof. Maxim Likhachev Sept. 2024 – Present

- Architecting a high-frequency **C++ motion planning framework (CTMP)** for contact-rich manipulation, targeting **deterministic (<50ms) planning times** for hook-latch disengagement tasks.
- Developing a **skill-extraction framework** that generalizes search-based plans into parameterized **motion skills**, enabling rapid adaptation to new hook-latch configurations without expensive global re-planning.
- Implementing a **hybrid controller** that dynamically switches between task-space Cartesian control and **active-joint subspace planning** (reducing effective search dimensions by ~60%) to navigate tight contact constraints.
- Building a high-fidelity **hardware-in-the-loop (HIL) benchmarking suite** using **ROS, MoveIt, and SAPIEN** to validate sim-to-real transfer on a UR10e manipulator under strict 10ms control loop limits.

16-832 Special Topics: Integrated Planning & Learning, CMU Pittsburgh, PA
Teaching Assistant | Advisor: Prof. Maxim Likhachev Dec. 2025 – Present

- Engineered a **high-reliability Python middleware** for ManiSkill/SAPIEN, abstracting low-level physics into a structured **planning-centric API** that standardizes state access and determinism for student research. **[code]**
- Architected a standalone **Multi-Agent Path Finding (MAPF) simulation engine** featuring custom collision checking, dataset ingestion, and real-time visualization to enable lightweight prototyping without heavy external simulators.**[code]**
- Developing an **automated evaluation pipeline** to benchmark planner performance, quantitatively analyzing **theoretical optimality guarantees** and cost-function convergence across diverse test suites.

PROJECTS

Iterative 2D–3D Inpainting for Single-Image 3D Reconstruction | CMU Sept. 2025 – Dec. 2025

- Architected a **self-correcting 3D reconstruction framework** that couples Latent Diffusion (2D inpainting) with Monocular Depth Estimation (Depth-Anything-V2) to synthesize and recover occluded geometry from single-view RGB inputs.
- Implemented a custom **3D inpainting algorithm** (Open3D, Poisson Reconstruction) that iteratively detects sparse point-density regions and repairs geometric noise, solving the "depth instability" problem inherent in generative lifting.
- Outperformed the **TripoSR baseline** on synthetic benchmarks, reducing Chamfer Distance by **55% (0.066 → 0.030)** and increasing Multi-View Consistency (MVCS) from **0.938 → 0.971**, demonstrating superior structural completeness.

Formation Control and Multi-Agent Pathfinding (MAPF) | CMU Sept. 2024 – Dec. 2024

- Engineered a high-performance **C++ formation control solver** (GIF-PIBT) that integrates **SVD-based rigid-body alignment** (using **Eigen**) into the PIBT heuristic, enabling real-time shape maintenance for large-scale swarms in obstacle-dense environments.
- Designed a hierarchical cost-routing framework that dynamically balances formation rigidity **vs.** goal progression, demonstrating superior shape preservation and routing stability **over** PIBT baselines across complex grid maps.

Dynamics Analysis of Model-Free RL Algorithms (NumPy-Only) | IIT Madras Jan. 2024 – May 2024

- **Engineered bare-metal implementations** of Dueling DQN and REINFORCE using **only NumPy and SymPy**, manually deriving and coding gradient updates to eliminate high-level framework overhead.
- Analyzed convergence failure modes in stochastic control tasks, identifying that while off-policy methods improved sample efficiency by 40%, they suffered from **high-variance instability** in noisy reward regimes compared to on-policy baselines.

SKILLS

Languages: C++17 (STL, Eigen), Python (PyTorch, NumPy, Pandas), MATLAB, Bash
Robotics & Simulation: ROS, MoveIt, Gazebo, ManiSkill (SAPIEN), URDF/Xacro, RViz
Planning & Control: Graph Search (A*, ARA*), Sampling-based (RRT*, PRM), Multi-Agent Path Finding (MAPF/PIBT), MPC
AI & Perception: Reinforcement Learning (PPO, DQN), Stable Diffusion, Transformers, Open3D, OpenCV, Point Clouds
Developer Tools: Linux, Docker, Git, CMake, GDB, CI/CD